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# =====
# Aliasing Artifact Demonstration
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import cv2
import numpy as np
import matplotlib.pyplot as plt
from google.colab import files

print("Upload an image")
uploaded = files.upload()

filename = list(uploaded.keys())[0]

# Read image
img = cv2.imread(filename)
img = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
```

Upload an image

ChatGPT I...4_19 PM.png

ChatGPT Image Jul 8, 2026, 08_24_19 PM.png(image/png) - 1809651 bytes, last modified: 08/07/2026 - 100% done
Saving ChatGPT Image Jul 8, 2026, 08_24_19 PM.png to ChatGPT Image Jul 8, 2026, 08_24_19 PM.png

```
# -----
# Original Image
# -----

# Create aliasing by aggressive downsampling
small = cv2.resize(img, (img.shape[1]//8, img.shape[0]//8),
                    interpolation=cv2.INTER_NEAREST)

aliased = cv2.resize(small,
                    (img.shape[1], img.shape[0]),
                    interpolation=cv2.INTER_NEAREST)
```

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# -----
# Corrective Approach
# Anti-aliasing filter
# -----

blur = cv2.GaussianBlur(img, (7,7), 0)

small_fixed = cv2.resize(blur,
                        (img.shape[1]//8, img.shape[0]//8),
                        interpolation=cv2.INTER_AREA)

corrected = cv2.resize(small_fixed,
                      (img.shape[1], img.shape[0]),
                      interpolation=cv2.INTER_LINEAR)
```

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# -----
# Display
# -----

plt.figure(figsize=(16,5))

plt.subplot(1,3,1)
plt.imshow(img)
plt.title("Original Image")
plt.axis("off")

plt.subplot(1,3,2)
plt.imshow(aliased)
```

```
plt.title("Aliasing Artifact")
plt.axis("off")

plt.subplot(1,3,3)
plt.imshow(corrected)
plt.title("Corrected using Anti-Aliasing")
plt.axis("off")

plt.show()
```

Original Image



Aliasing Artifact



Corrected using Anti-Aliasing



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# -----
# Explanation
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```
print("\nCause of Aliasing:")
print("- Sampling frequency is too low.")
print("- High-frequency image details fold into lower frequencies.")
print("- Produces jagged edges and moire patterns.")

print("\nQuantization Concept:")
print("- Low quantization levels reduce intensity precision.")
print("- This mainly introduces banding/noise rather than aliasing.")

print("\nCorrective Approach:")
print("- Apply Gaussian Blur before downsampling.")
print("- Increase sampling resolution.")
print("- Use proper interpolation (INTER_AREA) while resizing.")
```

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- High-frequency image details fold into lower frequencies.
- Produces jagged edges and moire patterns.

Quantization Concept:

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- This mainly introduces banding/noise rather than aliasing.

Corrective Approach:

- Apply Gaussian Blur before downsampling.
- Increase sampling resolution.
- Use proper interpolation (INTER_AREA) while resizing.

